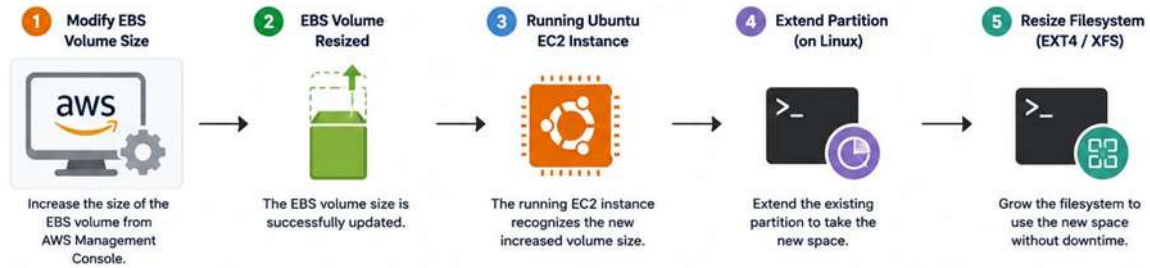
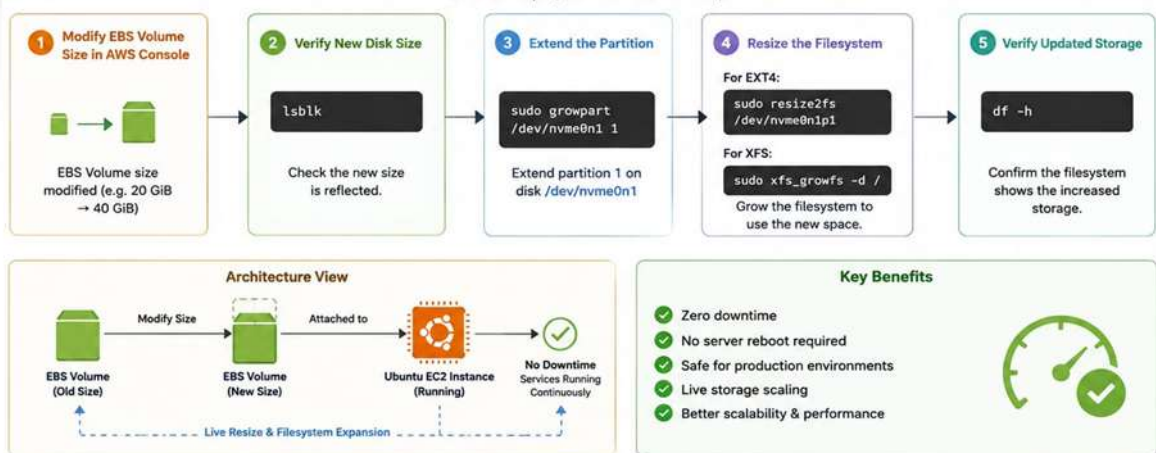


Resize EBS Volume on Ubuntu EC2 Without Downtime

Live Storage Expansion – No Reboot, No Downtime



Detailed Steps (Commands on Ubuntu)



-----A Mini Project By Shravani Poul-----

-----Steps To Resize EBS Volume Without Downtime-----

Step 1 :- Created An EC2 Instance With Default 20GB Volume Size

Created Volume 20Gb in Storage Section

Volumes (1) Info							Last updated less than a minute ago
Saved filter sets Choose filter set. Search							
Volume ID = vol-0df1105a0016ff2fc Clear filters							
	Name	Volume ID	Type	Size	IOPS	Throughput	
		vol-0df1105a0016ff2fc	gp3	20 GiB	3000	125	

```
ubuntu@ip-172-31-40-97:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0        7:0    0   48.4M  1 loop /snap/snapd/26382
loop1        7:1    0    74M   1 loop /snap/core22/2411
loop2        7:2    0   28.2M  1 loop /snap/amazon-ssm-agent/13009
nvme0n1      259:0    0    20G   0 disk 
├─nvme0n1p1  259:1    0   18.9G  0 part /
├─nvme0n1p13 259:2    0  1023M  0 part /boot
├─nvme0n1p14 259:3    0     4M   0 part 
└─nvme0n1p15 259:4    0   106M  0 part /boot/efi
```

Step 2 :- Modify The Volume in AWS Console

- Navigate to the EC2 Dashboard, select Volumes
- Select the volume, click Actions, and choose Modify Volumes
- Enter the new larger(30gb size) and click Modify
- Note: the volume state will change to "modifying" and then "optimizing" but remains available to the instance

Modify volume [Info](#)

Modify the type, size, and performance of an EBS volume.

Volume details

Volume ID
 vol-0df1105a0016ff2fc

Volume type [Info](#)
General Purpose SSD (gp3) ▼

Size (GiB) [Info](#)
20

Min: 1 GiB, Max: 65536 GiB.

Step 3 :- Extend the File System (Linux)

- SSH into the Instance
- Run lsblk to identify the block device name and confirm the new size is recognized. If there are partitions (e.g. nvme0n1p1) expand the partitions first using growpart:
- sudo growpart /dev/nvme0n1p1 1 (example)
- Extend the filesystem based on the filesystem type
- XFS: sudo xfs_growfs-d/
- EXT4: sudo resize2fs /dev/nvme0n1p1
- Verify with df -hT

```
ubuntu@ip-172-31-40-97: ~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0         7:0      0 48.4M  1 loop /snap/snapd/26382
loop1         7:1      0   74M  1 loop /snap/core22/2411
loop2         7:2      0 28.2M  1 loop /snap/amazon-ssm-agent/13009
nvme0n1       259:0     0   30G  0 disk 
├─nvme0n1p1   259:1     0 18.9G  0 part /
├─nvme0n1p13  259:2     0 1023M  0 part /boot
├─nvme0n1p14  259:3     0    4M  0 part 
└─nvme0n1p15  259:4     0 106M  0 part /boot/efi
```

Nvme0n1 Size :-30Gb

But if We check from df -ht in file system

```
ubuntu@ip-172-31-40-97: ~$ df -hT
Filesystem      Type      Size  Used Avail Use% Mounted on
/dev/root       ext4      19G   2.3G   16G  13% /
tmpfs           tmpfs     455M   0    455M   0% /dev/shm
tmpfs           tmpfs     182M  924K   181M   1% /run
efivarfs        efivarfs  128K   3.1K   120K   3% /sys/firmware/efi/efivars
tmpfs           tmpfs     455M   0    455M   0% /tmp
none            tmpfs     1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
none            tmpfs     1.0M   0    1.0M   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13 ext4      989M   96M   826M  11% /boot
/dev/nvme0n1p15 vfat      105M   6.3M   99M   7% /boot/efi
none            tmpfs     1.0M   0    1.0M   0% /run/credentials/systemd-networkd.service
none            tmpfs     1.0M   0    1.0M   0% /run/credentials/serial-getty@ttyS0.servic
none            tmpfs     1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           tmpfs     91M   8.0K   91M   1% /run/user/1000
```

EBS Volume Still Shows 20Gb (/dev/root)

As there are partitions of disk nvme0n1 (e.g nvme0n1p1), so expand the partition first using growpart: sudo growpart /dev/nvme0n1 1

```
ubuntu@ip-172-31-40-97: ~$ sudo growpart /dev/nvme0n1 1
CHANGED: partition=1 start=2324480 old: size=39618527 end=41943006 new: size=60590047 end=62914526
```

- Extend the file system based on the file system type: As we have ext4 file system so we'll use the command- `sudo resize2fs /dev/nvme0n1p1`
- Verify now the disk size using `df -hT` (shows file with human readable format). Now we can see the size of our volume `/dev/root` has resized from 20GB to 30GB

```
ubuntu@ip-172-31-40-97:~$ sudo resize2fs /dev/nvme0n1p1
resize2fs 1.47.2 (1-Jan-2025)
Filesystem at /dev/nvme0n1p1 is mounted on /; on-line resizing required
old_desc_blocks = 3, new_desc_blocks = 4
The filesystem on /dev/nvme0n1p1 is now 7573755 (4k) blocks long.
```

```
ubuntu@ip-172-31-40-97:~$ df -hT
```

Filesystem	Type	Size	Used	Avail	Use%	Mounted on
/dev/root	ext4	28G	2.3G	26G	9%	/
tmpfs	tmpfs	455M	0	455M	0%	/dev/shm
tmpfs	tmpfs	182M	924K	181M	1%	/run
efivarfs	efivarfs	128K	3.1K	120K	3%	/sys/firmware/efi/efivars
tmpfs	tmpfs	455M	0	455M	0%	/tmp
none	tmpfs	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
none	tmpfs	1.0M	0	1.0M	0%	/run/credentials/systemd-resolved.service
/dev/nvme0n1p13	ext4	989M	96M	826M	11%	/boot
/dev/nvme0n1p15	vfat	105M	6.3M	99M	7%	/boot/efi
none	tmpfs	1.0M	0	1.0M	0%	/run/credentials/systemd-networkd.service
none	tmpfs	1.0M	0	1.0M	0%	/run/credentials/serial-getty@ttyS0.service
none	tmpfs	1.0M	0	1.0M	0%	/run/credentials/getty@tty1.service
tmpfs	tmpfs	91M	8.0K	91M	1%	/run/user/1000